Effect of Overfitting on model parameter “W”

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\|W\|^2 = 4$$

Test

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
60	80	100

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 2.7 \\ 0.67 \\ 0.64 \end{pmatrix}$$

$$\|W\|^2 = 8.11$$

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
32.4	42.3	52.3

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 4 \\ 0.29 \\ 0.78 \end{pmatrix}$$

$$\|W\|^2 = 16.7$$

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
24.4	31.2	38